

3rd Subject

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DataBase Design

- a. Using a database from one or more different domain, show examples of strong and weak entities, optional and mandatory participation constraints.
- b. Consider the relation schema $R(A, B, C, D)$. Show which functional dependencies should hold in order to have A and B as Primary Key and relation R to be in the second normal form 2 NF.
- c. Consider the relation schema $R(A, B, C, D, E)$ with the next functional dependencies $AB \rightarrow C, C \rightarrow D, C \rightarrow E, D \rightarrow A, E \rightarrow B$
 - a. Find a key for R.
 - b. State the normal form for R and describe functional dependencies that violate higher NF.